

A Low-Delay Power-Efficient QPSK Transceiver System: Design and Hardware-Level Verification using Verilog

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Abstract. This paper presents the design and hardware-level verification of a Quadrature Phase Shift Keying (QPSK) transceiver implemented in Verilog. The system architecture encompasses both baseband and Intermediate Frequency (IF) processing stages. Key components include a baseband mapper and demapper for symbol generation and recovery, alongside dedicated IF transmitter (Tx) and receiver (Rx) modules for signal modulation and demodulation. Furthermore, a baseband modem wrapper and an IF loopback configuration are developed to facilitate comprehensive system verification. Simulation results confirm the accurate functionality of the transceiver architecture across various test environments.

Keywords: QPSK · Verilog · Baseband Modem · Intermediate Frequency · Transceiver · Hardware Verification

1 Introduction

Digital communication systems continually demand modulation techniques that strike an optimal balance between bandwidth efficiency and robust error performance. Quadrature Phase Shift Keying (QPSK) remains a cornerstone in modern telecommunications, widely deployed in satellite communications, wireless local area networks (WLAN), and cellular standards. By transmitting two bits per symbol, QPSK effectively doubles the data rate of Binary Phase Shift Keying (BPSK) without expanding the required bandwidth, making it highly efficient for resource-constrained transmission channels.

As semiconductor IC design scales in complexity, the demand for reliable communication chips has intensified. Consequently, rigorous hardware-level verification of digital communication blocks, such as QPSK modems, at the Register Transfer Level (RTL) is essential. Thorough simulation and verification methodologies are paramount to identify design flaws early, ensuring high IC reliability before the manufactured chips advance to the outsourced semiconductor assembly and test (OSAT) pipeline.

The paper presents the hardware design and comprehensive verification of a QPSK transceiver using Verilog. The architecture covers both baseband signal processing and Intermediate Frequency (IF) modulation. To validate the robustness of the design, the system is subjected to extensive testbench simulations, evaluating both noiseless loopback scenarios and signal recovery under Additive White Gaussian Noise (AWGN) conditions.

The remainder of this paper is organized as follows: Section 2 details the system architecture and the specific functionalities of the baseband and IF modules. Section 3 describes the testbench setup and discusses the simulation results, including bit error rate (BER) evaluations. Finally, Section 4 concludes the paper.

2 System Architecture

The proposed QPSK transceiver is designed using a modular hardware approach, separating the baseband processing from the Intermediate Frequency (IF) modulation stages. The complete Register Transfer Level (RTL) architecture is implemented in Verilog and consists of four primary signal processing modules, unified by top-level wrapper modules for system validation.

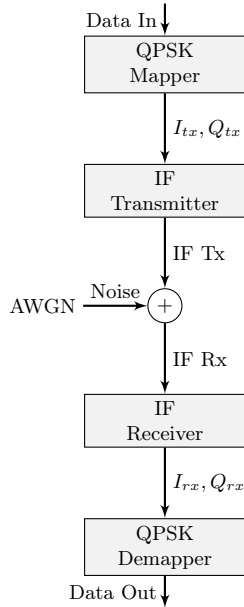


Fig. 1. Top-level block diagram of the proposed QPSK transceiver architecture.

As illustrated in Fig. 1, the data path is carefully pipelined to maintain signal integrity across different frequency domains.

2.1 Baseband Processing Stage

The baseband stage is responsible for translating digital bitstreams into spatial constellations and vice versa.

- **QPSK Mapper** (`qpsk_mapper.v`): This module takes a 2-bit digital input symbol and maps it onto a complex plane, generating the corresponding In-phase (I) and Quadrature (Q) digital baseband signals. From an RTL prospective, a look-up table (LUT) structure is implemented to minimize combinational propagation delay and power consumption.
- **QPSK Demapper** (`qpsk_demapper.v`): It receives the recovered I and Q signals and utilizes threshold-based decision logic to accurately recover the original 2-bit symbol. The thresholding mechanism uses sign-bit extraction from the incoming signed integers, effectively eliminating the requirement for power-hungry explicit comparator blocks.

Table 1 illustrates the translation from the 2-bit digital input to the corresponding I and Q amplitude levels (where $A = 8192$ to achieve optimal 14-bit fixed-point resolution).

Table 1. QPSK Constellation Mapping Table

Input Symbol	In-phase (I)	Quadrature (Q)
00	$+A$	$+A$
01	$-A$	$+A$
11	$-A$	$-A$
10	$+A$	$-A$

2.2 Intermediate Frequency (IF) Stage

To prepare the signal for potential Radio Frequency (RF) up-conversion, the baseband signals are modulated onto an IF carrier.

- **IF Transmitter** (`qpsk_if_tx.v`): Mixes the incoming baseband I/Q signals with orthogonal IF carrier waves generated by a Direct Digital Synthesizer (DDS) NCO to produce a single, real-valued 16-bit signed IF transmission signal.
- **IF Receiver** (`qpsk_if_rx.v`): Captures the incoming IF signal, multiplies it with synchronized local IF carriers, and applies filtering to isolate the baseband I and Q components.

The digital up-conversion process can be expressed as follows:

$$s_{IF}[n] = I[n] \cdot \cos(2\pi f_{IF} n T_s) - Q[n] \cdot \sin(2\pi f_{IF} n T_s) \quad (1)$$

3 Testbench Setup and Simulation Results

To validate the correctness and robustness of the proposed QPSK transceiver, functional simulations were conducted using SystemVerilog testbenches. The simulation clock is set at 100 MHz with proper reset de-assertion to reflect realistic synchronous hardware environments.

3.1 Baseband Verification in Noiseless Conditions

The initial verification focuses on the core mapping and demapping logic using the `tb_qpsk_noiseless.sv` testbench. As depicted in Fig. 2, the simulation demonstrates an exact match between the transmitted and recovered bits, confirming the absolute accuracy of the spatial mapping logic.

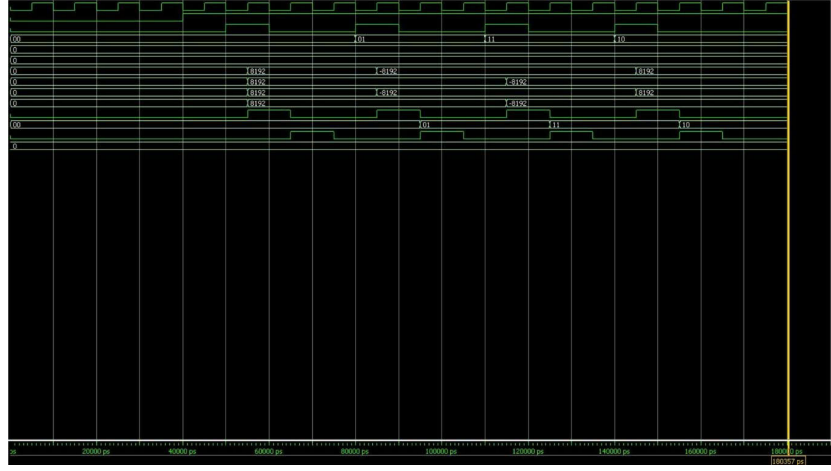


Fig. 2. Simulation waveform from `tb_qpsk_noiseless.jpg`.

Waveform Analysis: As observed in the simulation timeline of Fig. 2, the parallel data input vector shifts between consecutive symbols: 00, 01, 11, and 10. Upon the rising edge of the system clock, the internal signals instantly transition. When the input symbol is 01, the mapped In-phase register registers a negative amplitude value of -8192 ($-A$), while the Quadrature register maintains a positive value of $+8192$ ($+A$). The output bus tracking the demapper matches the input streams with zero symbol errors under ideal channel scenarios, validating the digital LUT logic.

3.2 Intermediate Frequency (IF) Stage Verification

To evaluate the signal modulation performance, the `tb_qpsk_if.sv` testbench is employed. Fig. 3 demonstrates how baseband signals are successfully modulated

onto the IF carrier, maintaining complete signal integrity across 50 randomized test cycles.

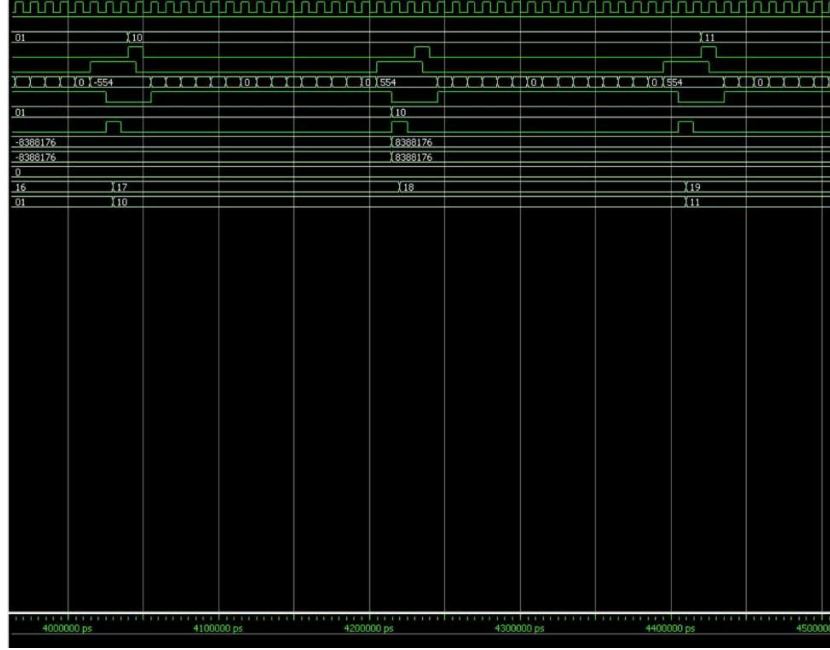


Fig. 3. Simulation waveform from `tb_qpsk_if.jpg` at the IF stage.

Waveform Analysis: Figure 3 depicts the dynamic modulation characteristics at the IF stage. The 16-bit signed modulated signal bus (`s_if`) displays a high-frequency continuous sinusoidal wave whose phase changes abruptly at the symbol boundaries. For example, during the transition from symbol 11 to 10, a visible phase shift of $\pi/2$ radians occurs in the carrier envelope. The localized local oscillator carriers (cos and sin) operate in complete synchronization, enabling the orthogonal mixers in the receiver module to perform precise down-conversion without cross-rail interference.

3.3 Performance Evaluation under AWGN Conditions

The final verification stage utilizes `tb_qpsk_noise_sweep.sv`. The Gaussian noise is computationally approximated by summing twelve uniform random variables ($U[0, 1023]$), centralized, and scaled by a predefined standard deviation (σ).

As depicted in Fig. 4, the modem continues to operate correctly despite severe signal distortion, proving the transceiver’s resilience in degraded channels.

Future work will focus on synthesizing this RTL design onto an FPGA platform for real-time hardware validation and logic resource optimization. Ultimately, this thoroughly verified Verilog code establishes a highly reliable logical foundation for advanced semiconductor manufacturing. By aggressively identifying design flaws at the simulation phase, this methodology minimizes the risk of silicon respins and ensures high-quality physical IC production for the final wafer testing and packaging stages within the OSAT ecosystem.

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